Elliott Ashby

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Education

University of Southampton 2021 — 2026
Master of Physics (MPhys) *Southampton, UK*
- Dissertation: *Quantum Simulation of the Schwinger Model on a Ring* — simulating particle scattering on quantum computers.
- Implemented ADAPT-VQD spectra, wave packets and Trotter evolution in Qiskit, with MPS for larger lattices.

Experience

Chief Technology Officer Nov 2026 — Present
Anerri Bio Limited *London, UK*
- Lead technical direction and ML for a biotech designing peptide-based immunotherapies for food allergies.

Analyst Aug 2026 — Oct 2026
Perscitus LLP *London, UK*
- Developed early ML approaches for peptide immunotherapy before moving to Anerri Bio on its formation.

Intern May 2024 — Sep 2024
Transreport Japan K.K. *Tokyo / Osaka, Japan*
- Supported expansion into the Japanese rail market: industry research, client engagement and EN–JP translation.
- Contributed to market-entry strategy, localisation requirements and stakeholder engagement with Japanese rail operators.

Database Developer and Analyst 2019 — Aug 2022
Perscitus LLP *London, UK*
- Built the grants database for a medical research charity, tracking £53m across 170+ clinical trials (jmcharitytrust.gg).
- Replaced manual tracking with a desktop app, saving the administrator significant time and making reporting far faster.

Personal Tutor 2021 — Present
Self-employed
- Provide private tuition in A-level Physics and GCSE Physics and Chemistry.

Scout Leader 2017 — Present
The Scout Association *Sevenoaks, UK*
- Volunteer leader for young people aged 10–14, with training in safeguarding and first aid.

Selected Projects

Creator, onon ononreader.com
- Keyboard-driven document reader in Go on MuPDF with Lua scripting; released for Linux, macOS and Windows.

Creator, Noev GitHub/Aethar01/noev
- Hybrid classical–quantum programming language with ownership-typed qubits and automatic circuit extraction.

Creator, ketrs GitHub/Aethar01/ketrs
- Rust quantum simulator with Python and GPU support; 300× faster than Qiskit Aer on dynamic circuits at 20 qubits.

Skills and Qualifications

Programming: Python, Rust, Go, C, SQL, Bash
Machine Learning: PyTorch, TensorFlow, Hugging Face Transformers, safetensors, ESM protein language models
Scientific Computing: NumPy, SciPy, Pandas, Matplotlib, Qiskit, statistics and data analysis
AI-Assisted Development: Claude Code, OpenAI Codex, LLM APIs, Model Context Protocol (MCP), agentic workflows
Infrastructure and Tools: Linux, Docker, Git, SQLite, Azure SQL Database, FastAPI, NGINX, ZFS, LaTeX/Typst
Languages: Japanese (fluent)
Qualifications: Duke of Edinburgh's Gold Award, Enhanced DBS Certificate, UK Driving Licence
Music: Grade 8 Clarinet, Grade 8 Saxophone, Grade 6 Piano, Grade 5 Music Theory

Hobbies and Interests

Technology: Computer hardware, Linux, open source (Firefox contributor)
Science: Quantum mechanics, quantum computing
Music: Clarinet, saxophone, piano, bass
Other: Climbing, bouldering, tabletop role-playing game play and design